

Representation Collapse Is Not the Bottleneck: A Negative Result and Diagnosis for Vision-Based Quadrotor Hover

Draft v0.3 — 2026-06-19. Target: ICRA / robot-learning workshop (venue TBD). **Simulation-only; no real-robot claim.** This draft is assembled from the frozen-protocol leaderboard and the Phase 3 diagnostic reports; every number is reproducible from the cited script/artifact.

Abstract

Diffusion / flow-matching visual policies are prone to *representation collapse* when fine-tuned end-to-end, and **Dispersive Loss** — a contrastive regulariser that repels intermediate features — has been proposed to prevent it and thereby improve high-frequency visual control. We pre-registered and tested this hypothesis on a vision-based quadrotor hover task (monocular 64×64 FPV + IMU, flow-matching policy, 50 Hz closed loop). Under a **frozen evaluation protocol** (paired initial conditions, conditional-on-survival precision, bootstrap CIs, across-seed mean $\pm$ std, a *measured* state-based oracle), a **2×2 ablation** (Dispersive $\times$ end-to-end encoder, three seeds per cell) finds that **Dispersive Loss yields no survival or task-precision gain above seed noise** (+1.1 pp Tier-1 within a 4.2 pp pooled std), and with a frozen encoder it is **byte-identical** to its absence. Reaching even this null verdict required a protocol hardened against **single-seed noise**: our own from-pixels PPO baseline swings between a 0 % and 47 % Tier-1 pass-rate on the training seed alone, so we report every retrained model as an **across-seed mean $\pm$ std** rather than a single leaderboard row. We then diagnose *why*: (i) Dispersive does not cure collapse — it **games its own objective**, inflating feature norm $\sim 287\times$ while the intrinsic rank gets *worse* (effective rank 9 $\rightarrow$ 2), and closed-loop survival is **decoupled** from that rank (a $15\times$ rank swing leaves survival flat); (ii) the only real survival mover is a transfer/conditioning/recovery **recipe**, not Dispersive; (iii) precision is gated neither by representation nor by sensing. Although the 64×64 FPV cannot encode metric range beyond ~ 2 m, a renderer gate shows that loss is a fixable target artifact (a non-saturating target restores it at 64 px), and a positive-control **intervention** settles it: *handing the policy the oracle metric position error barely moves precision* ($\sim 36\times$ the state oracle) and a richer cue **collapses survival** — so precision is capped by the absence of a learned far-range recovery behaviour (a coverage / teacher-competence gap), not by the observation channel. We conclude that for this class of task neither representation collapse nor sensing is the binding constraint. We release the protocol, the intervention, and all diagnostics.

1. Introduction

End-to-end visual control with generative policies (diffusion [3], flow matching [4]) has advanced rapidly, but a recurring failure mode is *representation collapse*: when the visual encoder is trained jointly with the action head, its features can degenerate onto a low-dimensional subspace, losing the information the controller needs. **Dispersive Loss** [13] addresses this by adding a contrastive term that repels a batch's intermediate features, and has been reported to improve generative-policy control [14].

We asked a simple, falsifiable question on a concrete task — **vision-based quadrotor hover** — does Dispersive Loss improve closed-loop control, and does it need an end-to-end-trainable encoder to do so? The task is a good stress test: the policy runs at 50 Hz from a monocular 64×64 FPV stream plus IMU, with a flow-matching action head; covariate shift and feature collapse are both plausible failure modes.

Our answer is **no**, and the value of the paper is the *diagnosis* of why. We make three contributions:

1. **A frozen evaluation protocol** for visual hover that is robust to the short-survival *and* single-seed artifacts that silently mislead precision metrics and leaderboard rows for policies that crash early or train unstably (paired seeds, conditional-on-survival precision, bootstrap CIs, across-seed mean $\pm$ std, and a *measured* rather than assumed oracle).
2. **A pre-registered 2×2 falsification** (three seeds per cell) of the Dispersive-Loss hypothesis, including a byte-identical no-op control.
3. **A mechanistic diagnosis** separating the three candidate bottlenecks (representation, data coverage, sensing) and localising the real one — including a positive-control *intervention* that hands the policy the oracle sensing

signal and shows it does **not** restore precision, isolating the teacher's far-range incompetence as the binding constraint.

This is a negative result, but a constructive one: it redirects effort from representation regularisation to the observation model.

2. Related Work

Generative visuomotor policies. Denoising diffusion models [1] and their deterministic DDIM sampler [2] made expressive, multimodal generative modelling practical, and Diffusion Policy [3] adapted this to visuomotor imitation by casting action prediction as conditional action diffusion. Flow matching [4] and rectified flow [5] provide a simulation-free, ODE-based alternative that trains a velocity field along straight transport paths; both have since been used as robot-control policies — e.g. the π_0 vision-language-action flow model [6] and the Riemannian flow-matching policy for motion learning [7]. Our policy is a flow-matching action head over a fused vision-IMU conditioning vector, placing it in this family.

RL fine-tuning of generative policies. Behaviour-cloned generative policies inherit the demonstrator's covariate shift, motivating RL fine-tuning. DPPO [8] treats the multi-step denoising chain as an MDP and fine-tunes a pre-trained diffusion policy with PPO, reporting stable, on-manifold exploration for continuous control. ReinFlow [9] extends this to flow policies by injecting *learnable, bounded noise* into the otherwise deterministic flow ODE, turning it into a discrete-time SDE with closed-form transition probabilities so that exact log-likelihoods — and hence policy gradients — are available at very few denoising steps; Q-score matching [12] is a related value-based route. This interactive, continuous-control setting should be distinguished from RL fine-tuning of *image-generation* diffusion models (DDPO [10]; DPOK / Fan et al. [11]), which optimise a non-interactive sampler against a reward and do not face closed-loop dynamics. Our RL stage (ReinFlow with positive-advantage masking) sits squarely in the former group.

Representation collapse and dispersive regularization. Generative models trained with a pure regression objective lack explicit pressure on their internal features, which can collapse semantically distinct observations onto near-identical representations. Dispersive Loss [13] counters this with a "contrastive loss without positive pairs" — keeping only the repulsion term so that a batch's intermediate features spread out, with no positive pairs, augmentation, or extra encoder. D²PPO [14] imports this regulariser into diffusion-policy *pre-training*, reporting large manipulation gains ($\approx +22.7\%$ success after pre-training and $+26.1\%$ after PPO fine-tuning on RoboMimic) attributed to relieving representation collapse — the precise claim our task is designed to test. Related anti-collapse mechanisms include representation alignment to a pretrained encoder (REPA [15]) and the classic variance/covariance and redundancy-reduction objectives of VICReg [16] and Barlow Twins [17]; concurrent work also pairs dispersive-style regularisation with MeanFlow for one-step manipulation (DM1 [18], MP1 [19]). We adopt Dispersive Loss exactly as specified [13, 14] and find, on our task, no closed-loop benefit (§5–§6): against D²PPO's reported $+22.7\%$ / $+26.1\%$ manipulation-success gains on RoboMimic [14], we measure a $+1.1$ pp Tier-1 change that sits inside the 4.2 pp across-seed std, with survival and precision unmoved — and, with a frozen encoder, a byte-identical no-op. We attribute the gap to the setting (interactive 50 Hz closed-loop flight vs. quasi-static manipulation pre-training) and diagnose it mechanistically in §6.

Learning-based vision quadrotor control. End-to-end learned controllers have pushed agile flight well past classical sense-map-plan pipelines: Loquercio et al. [20] fly a quadrotor through cluttered wild environments at high speed from onboard vision, trained in simulation via *privileged learning* (imitating a state-privileged teacher), and the Swift system [21] beats human champions at drone racing with deep RL on a vision-based policy. A common thread is decoupling perception from control via a privileged teacher — the same paradigm as our state-based PPO oracle (§3). Relative to these agility/navigation results, our task isolates a different axis: closed-loop *hover precision* under an information-poor monocular observation — where, by intervention (§6.3), the binding constraint turns out to be the privileged teacher's own incompetence in the operating regime, not the sensing channel.

3. Task, Policy, and the Dispersive Mechanism

Task. Hover a quadrotor at a target. The simulator is a 6-DOF rigid body (200 Hz RK4 inner loop, INDI attitude control) with a 50 Hz outer control loop. The agent observes a 2-frame stack of 64×64 RGB synthetic FPV images and a

6-D IMU vector (gyro + specific force); it outputs a short sequence of collective-thrust/body-rate (CTBR) commands, executing the first and re-observing (receding horizon).

Policy (flow matching, v5). A CNN vision encoder produces a pooled feature $\text{vis_pooled} \in \mathbb{R}^{256}$ and a spatial map $\mathbb{R}^{256 \times 4 \times 4}$. An IMU MLP produces $\text{imu_feat} \in \mathbb{R}^{512}$. An IMU-to-vision cross-attention (IMU query, spatial keys/values) produces $\text{attended} \in \mathbb{R}^{256}$. The flow network conditions on $\text{global_cond} = [\text{attended}; \text{imu_feat}] \in \mathbb{R}^{768}$. A training-only auxiliary head predicts the 15-D privileged state from vis_pooled .

Dispersive Loss. Applied to vis_pooled : $\mathcal{L}_{\text{disp}} = -\text{mean}_{\{i \neq j\}} \log(\|x_i - x_j\| + \epsilon)$, added with weight $\lambda=0.05$. It pushes the batch's pooled features apart. **Note (important for §6.1):** vis_pooled feeds only the auxiliary state head; the *action* path conditions on attended (from the spatial map) and imu_feat . Dispersive therefore reaches the action path only indirectly, through the shared convolutional trunk.

Oracle. A state-based PPO controller reading the privileged 15-D state is the performance upper bound.

4. Frozen Evaluation Protocol

The project's metric had changed three times (RMSE $\rightarrow$ linear-clip hierarchical $\rightarrow$ exp-decay hierarchical), making cross-run numbers incomparable; a **short-survival artifact** kept inflating precision for policies that crash early (a policy that dies at step 24 was credited with a "precise" 0.6 m hover from its few alive steps); and, as we demonstrate below with our own PPO baseline, **single-seed evaluation is itself unreliable** — a collapse-prone policy can swing from Tier-1 0 % to 47 % on the training seed alone. We freeze one protocol (`evaluate_frozen_p0.py`) and do not change it:

- **Paired initial conditions.** Episode i uses $\text{seed} = 12345 + i$ to seed the env, the global RNG (visual domain randomisation), and torch (flow noise), so every model sees byte-identical initial conditions — a paired comparison.
- **Primary axes: survival and Tier-1 pass-rate** (fraction of 30 episodes flying $\geq 250/500$ steps). These are robust to the short-survival artifact.
- **Conditional precision.** Integrated absolute error (IAE) of position is reported **only over episodes that survived ≥ 250 steps** (`cond-IAE` , with its support `n_cond`); the naive all-episode IAE is reported but flagged.
- **Bootstrap 95% CIs** over episodes.
- **Across-seed mean $\pm$ std** for every model we *retrain* in this work — Table 1's P1 baselines and Table 2's 2×2 cells — over three training seeds. The legacy flow-policy checkpoints predate this work and are single-seed; that is a stated limitation, and the seed sensitivity we document below is exactly why a single training run is not a safe basis for a leaderboard row.
- **Measured oracle.** The state-based PPO oracle is rolled through the *same* protocol; its composite score (**0.9668**, 100 % survival, 0.068 m hover IAE) is the normaliser (`%Oracle`), replacing a previously hard-coded constant.

Under this protocol, **three single-seed claims fail to survive**. Two come from the prior leaderboard: "H4 = SOTA" (true only on a multiplicative composite that rewards *die-early-but-precise*; H4 survives 40.8 %, Tier-1 13.3 %) and "RL gives 51 % precision gain" (a short-survival artifact: its conditional IAE is the *worst* of the group, 3.34 m, on `n_cond` = 4/30). The third we generate ourselves and catch with the seed protocol: on a single seed our PPO-from-pixels baseline "uniformly collapses" (Tier-1 0 %), but over three seeds it is only *collapse-prone* — a high-variance Tier-1 of **15.6 $\pm$ 22.0 %**, with one seed reaching 47 % — so the single-seed verdict simultaneously over- and under-states the behaviour. The lesson is structural, not incidental: on this task one training seed is not a safe basis for a leaderboard row, which is why every model we retrain is reported as a seed mean $\pm$ std. **No vision policy exceeds ~ 17 % of oracle; none is deployable.** (Table 1.)

Table 1 — Frozen leaderboard (flow policies: `evaluation_results/frozen_p0_leaderboard.json` ; P1 baselines ^B, 3 seeds: `evaluation_results/baselines_frozen_leaderboard.json` , `evaluation_results/baselines_frozen_ppopx.json` , per-seed `...{bc,ppopx}_s{1,2}.json` , aggregate `...seeds_aggregate.json`).

Model	Score	Survive	Tier-1	cond-IAE (n)	%Oracle
PPO Oracle (state)	0.967 [.966,.967]	100 %	100 %	0.068 m (30)	100 %
H4_BC	0.168 [.154,.182]	40.8 %	13.3 %	2.520 m (4) Δ	17.4 %
v5_RL_best	0.129 [.117,.139]	28.8 %	13.3 %	3.340 m (4) Δ	13.3 %
v5_BC	0.124 [.107,.143]	55.9 %	70.0 %	2.724 m (21)	12.8 %
PPO-from-pixels ^B	0.114 $\pm$ 0.017	30.1 $\pm$ 15.6 %	15.6 $\pm$ 22.0 %	2.83 m (1/3) $\Delta\Delta$	11.8 $\pm$ 1.8 %
Joint_E2E_v5	0.106 [.095,.118]	62.2 %	80.0 %	3.077 m (24)	11.0 %
BC-vision-only ^B	0.089 $\pm$ 0.018	54.7 $\pm$ 3.4 %	61.1 $\pm$ 13.4 %	3.30 $\pm$ 0.36 m (18)	9.2 $\pm$ 1.8 %

Δ n_cond = 4: cond-IAE unreliable; the low all-IAE is a short-survival artifact. $\Delta\Delta$ PPO-from-pixels is collapse-prone, not uniformly collapsed: 2 of 3 seeds survive 0/30 past 250 steps (n_cond = 0, cond-IAE undefined; their low all-IAE 0.70–0.87 m is a pure short-survival artifact, crashing $\sim$ 75 steps), while 1 seed reaches Tier-1 46.7 % (cond-IAE 2.83 m). The cond-IAE shown is that single surviving seed (1/3). ^B P1 baseline (this work): **mean $\pm$ std over 3 seeds {0,1,2}** (population std, matching Table 2); the bracketed [.,.] on the other rows are bootstrap 95 % CIs over a single checkpoint's episodes. BC-vision-only = no IMU, no flow (plain vision $\rightarrow$ action MLP); PPO-from-pixels = end-to-end RL from pixels (1 M steps, CTBR).

The two P1 baselines (3 seeds each) sit within the flow-policy band and reinforce the diagnosis. A plain vision $\rightarrow$ action regressor (BC-vision-only: no IMU, no flow) already reaches **9.2 $\pm$ 1.8 % of oracle at Tier-1 61.1 $\pm$ 13.4 %** — *on par with* the IMU-fused flow policies v5_BC (12.8 %, Tier-1 70 %) and Joint_E2E_v5 (11.0 %, Tier-1 80 %), well within the wide across-seed spread. Flow modelling and IMU fusion thus buy little in survival on this task: **policy sophistication is not the bottleneck.** The other baseline, PPO-from-pixels, is the collapse-prone case dissected above; here it doubles as a clean short-survival-artifact demonstration. On its two collapsed seeds the deceptively low *all*-IAE (0.70–0.87 m) is a textbook instance of what the §4 protocol is built to expose — they "look precise" only because they crash ($\sim$ 75 steps) before they can drift — whereas averaging over seeds raises the all-IAE to 1.41 $\pm$ 0.89 m once the surviving seed's drift is counted. Both baselines stay far below the 17 % ceiling; none is deployable.

One training seed is not a safe leaderboard row (P1 baselines, 3 seeds each)

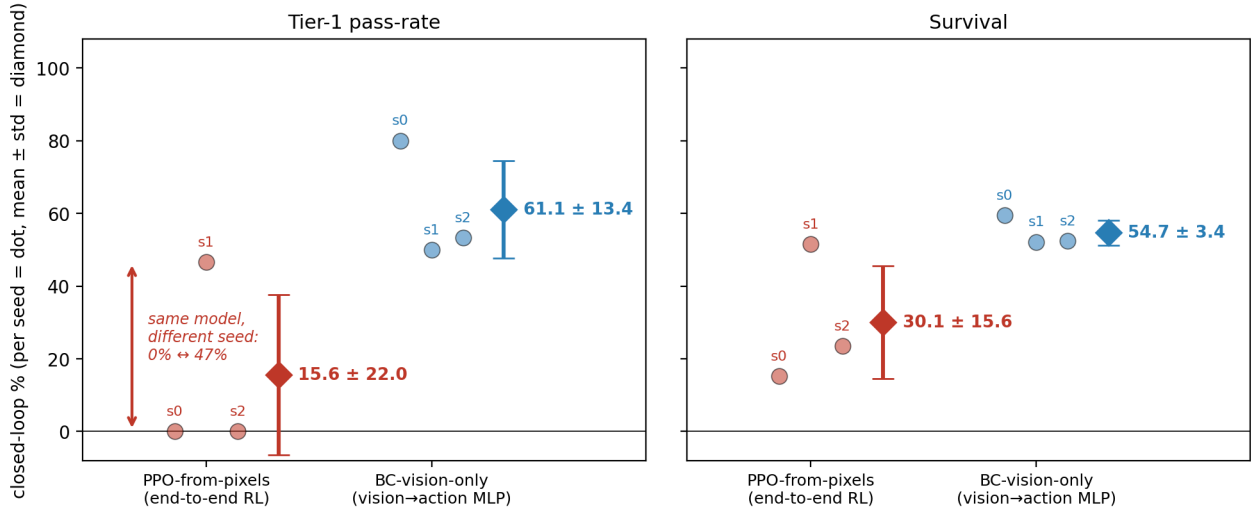


Figure 3. Per-seed Tier-1 pass-rate (left) and survival (right) for the two P1 baselines, three training seeds each (dots, labelled s0–s2), with the across-seed mean $\pm$ std (diamond). PPO-from-pixels swings from 0 % (seeds 0, 2 — would read as "uniformly collapsed") to 47 % (seed 1) on Tier-1 from the training seed alone (15.6 $\pm$ 22.0 %), so any single-seed leaderboard row simultaneously over- and under-states it; BC-vision-only is comparatively stable (61.1 $\pm$ 13.4 %). This seed sensitivity is why every retrained model in this work is reported as a three-seed mean $\pm$ std (scripts/make_paper_figures.py, evaluation_results/baselines_frozen_seeds_aggregate.json).

5. The Core Ablation: Dispersive × E2E (2×2, 3 seeds)

We vary two factors, holding everything else fixed (H4-transfer init, 500 hover + 500 recovery demos, 80 epochs, lr 1e-4, batch 256, task-conditioned):

- **Dispersive**: $\lambda=0.05$ (ON) vs 0.0 (OFF), on `vis_pooled`.
- **E2E**: vision encoder trainable (ON) vs frozen at the transferred init (OFF).

Mechanistic prediction (pre-registered). Dispersive acts on `vis_pooled`; with a frozen encoder its gradient touches no trainable parameter, so **D1E0 must equal DoE0**. The decisive comparison is therefore **D1E1 vs DoE1** (Dispersive ON vs OFF, both E2E).

Table 2 — 2×2 ablation (`evaluate_p2_ablation.py` , 3 seeds, frozen protocol).

Cell	Disp / E2E	Tier-1 (mean±std)	Survival	cond-IAE	%Oracle
DoE0	OFF / frozen	87.8 ± 3.1	66.1 ± 3.7	2.93 m	13.4 %
D1E0	ON / frozen	87.8 ± 3.1	66.1 ± 3.7	2.93 m	13.4 %
DoE1	OFF / E2E	92.2 ± 3.1	65.0 ± 2.8	2.91 m	12.4 %
D1E1	ON / E2E	93.3 ± 2.7	65.0 ± 2.5	2.81 m	13.2 %

Findings. - **Dispersive is not supported.** D1E1 vs DoE1: Tier-1 +1.1 pp inside the 4.2 pp pooled across-seed std; survival identical (65.0); precision unchanged. - **Byte-identical no-op.** D1E0 $\equiv$ DoE0: the trained `best_model.pt` files are **MD5-identical across all three seeds** — direct confirmation that, with a frozen encoder, "Dispersive ON" is bit-for-bit "Dispersive OFF". - The decision rule (Dispersive supported iff D1E1 beats DoE1 on Tier-1/survival by more than the pooled std) returns **NOT supported**.

All four cells beat the prior frontier Joint_E2E_v5 (Tier-1 80 %), but §6.2 shows that gain is the *recipe*, not Dispersive. Precision is unmoved (cond-IAE 2.8–2.9 m, ~13 % oracle): **nothing is deployable**.

6. Diagnosis: Which Bottleneck?

Three candidate explanations for the flat result: representation (the thing Dispersive targets), data coverage, or sensing. We test the first two with observation-only measurements (no retraining), and the third with both a renderer gate and a positive-control **intervention** that hands the policy the very signal it is presumed to lack (§6.3).

6.1 Representation: Dispersive games its objective; survival $\perp$ rank

We push a fixed 4000-image hover+recovery batch through each checkpoint's encoder and measure the geometry of `vis_pooled` (D=256). (`measure_feature_collapse.py` .)

Table 3 — feature geometry (mean over 3 seeds).

Cell	eff_rank	dims for 99% var	feat_norm	L_disp	mean cos
DoE0 / D1E0 (frozen)	30.3	91	1.80	−0.81	0.071
DoE1 (E2E)	9.0	45	11.42	−1.23	0.956
D1E1 (E2E + Disp)	2.0	2	3281	−8.14	0.013

Three things follow. **(i)** The frozen no-op is confirmed at the feature level (D1E0 $\equiv$ DoE0 to 4 d.p.). **(ii)** Collapse is *real*: naive E2E (DoE1) drops effective rank 30→9 and drives mean pairwise cosine to 0.96 (features nearly colinear) — there *was* something for Dispersive to fix. **(iii)** Dispersive does not fix it; it **games its objective**. It minimises `L_disp` (−1.23→−8.14) and inflates pairwise distance ~1270×, but does so purely by inflating the feature norm ~287× (11.4→3281); the *intrinsic* dimensionality gets **worse** (effective rank 9→2; 99.8 % of variance on 2 dims). The "healthy-looking" low cosine (0.013) is a 2-D-at-huge-radius artifact, not high- dimensional spread. Because `vis_pooled` feeds only the auxiliary head (§3), the cheapest way to satisfy the distance objective is to inflate the pooled-specific projection, leaving the action-relevant spatial pathway comparatively untouched.

The decisive plot is rank vs survival (Figure 1). Effective rank swings **15×** (30→9→2) while closed-loop survival stays flat (66.1→65.0→65.0 %, within ~3 pp std). **Survival is decoupled from vis_pooled rank.** Representation collapse — neither preventing it (DoEo rank 30) nor worsening it (D1E1 rank 2) — moves the closed-loop outcome. This is the strongest statement of the result: representation collapse is *not* the binding constraint.

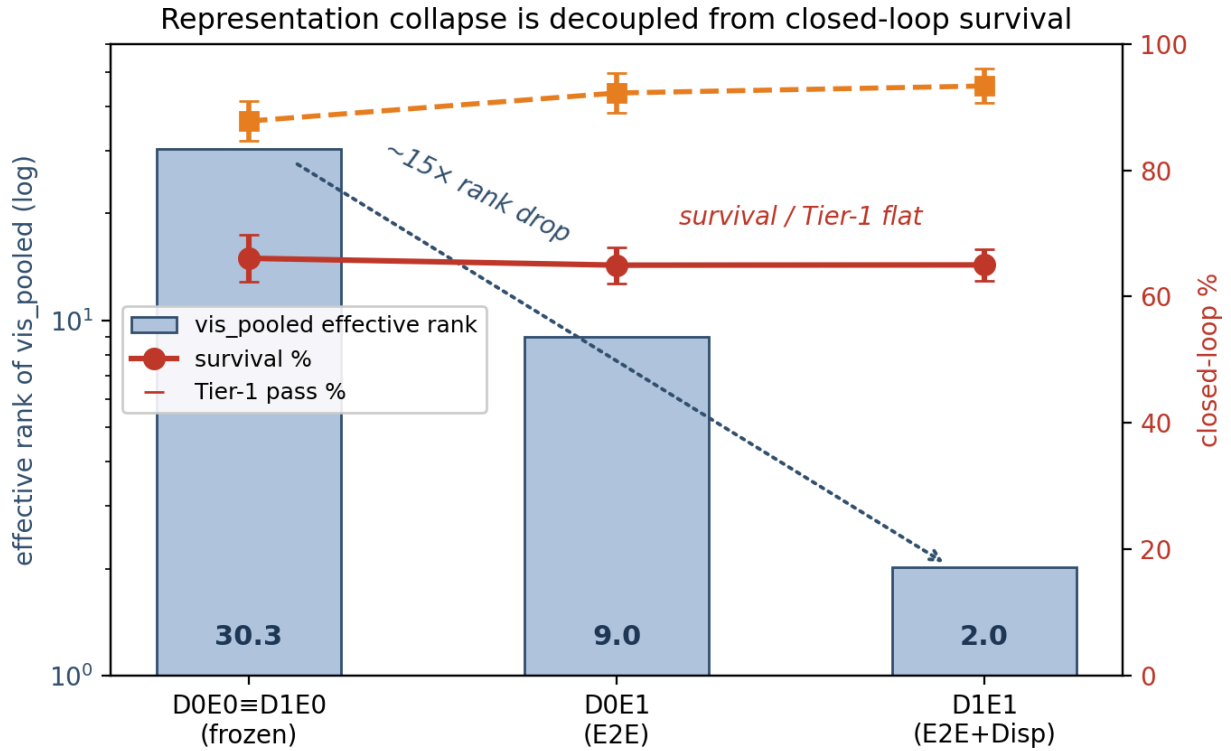


Figure 1. `vis_pooled` effective rank (bars, log axis) collapses ~15× across the 2×2 cells (30.3 frozen → 9.0 E2E → 2.0 E2E+Dispersive), while closed-loop survival and Tier-1 pass-rate (lines, right axis; error bars = across-seed std) stay flat. Survival is decoupled from the representation geometry Dispersive targets.

6.2 The only survival mover is the recipe, not Dispersive (or E2E)

Decomposing the gain over the prior frontier as controlled steps (`experiment_report_survival_movers.md`):

Step	Δ Tier-1	Δ Survival	cond-IAE
prior frontier Joint_E2E_v5 → recipe (frozen cell DoEo)	+7.8 pp	+3.9 pp	2.93 m
+ E2E (DoEo → DoE1)	+4.4 pp	-1.1 pp	2.91 m
+ Dispersive (DoE1 → D1E1)	+1.1 pp (€ noise)	0.0	2.81 m

The dominant lift is the **recipe** (H4-transfer init + task-conditioning + recovery mix), and it holds with a **frozen** encoder — so it is not from E2E. E2E adds a small Tier-1-only bump that neither extends survival nor improves precision. Dispersive adds nothing. **Mover ranking: recipe ≫ E2E ≫ Dispersive ≈ 0.**

6.3 Precision is coverage/teacher-competence-limited, not sensing-limited

cond-IAE is ~2.8 m (~13 % oracle) for *every* configuration. Why won't precision move?

Coverage probe (`measure_ood_coverage.py`). Comparing position-error magnitude in the BC data vs the closed-loop steady state: training is overwhelmingly at-target (95.8 % of timesteps within 0.2 m; p99 = 0.62 m), while the surviving policy spends ~90 % of its steady state beyond 1 m and 46 % beyond 3 m. **97.3 %** of steady-state samples lie above the training p99; **25 %** above the training maximum (zero-coverage). The policy lives where it has ~no labels — a real coverage gap, *consistent* with either a data limit or a sensing limit.

Is the coverage gap data-fixable, and does the FPV even encode range? Three observation-only checks bear on it: 1. **Widening the recovery init does not change coverage.** The env anchors `target = init_pos` for hover, so a wider `--pos-range` just relocates the hover point; a freshly collected ±3 m dataset has *identical* position-error

coverage to the ± 1 m one (both 0.4 % of timesteps beyond 1 m). 2. **The expert itself cannot recover from > 2 m offset** (target $\neq$ init): 1 m and 2 m recover to 0.066 m (20/20), but 3 m crashes (0/20). Even a correct offset collection tops out at ~ 2 m — below the precision-limiting regime. 3. **The FPV image does not encode metric range** (`measure_image_distance_info.py`; **Figure 2**). The only range cue is the target crosshair size, $\text{size} = \max(2, \min(6, \lceil 6/(d+0.5) \rceil + \text{dr}))$. With domain randomisation, adjacent- distance separability is $d' < 0.2$ everywhere, and a linear decode of distance from the image has $R^2 = 0.41$ near (< 1 m) but only **0.12 far** (≥ 1.5 m). With DR off (noiseless ceiling) the size **saturates at 2 px for all $d \geq 2$ m**, so renders at 2.0, 2.5, and 3.0 m are **byte-identical** ($d' = 0$). The crosshair *position* encodes only normalised direction (range-invariant).

The policy's 2.83 m steady drift sits exactly where the observation carries **no recoverable metric range** (Figure 2). It is tempting to stop here and call precision *information-gated by the observation model* — the conclusion of an earlier draft. But a measurement that the FPV does not encode range does not establish that the missing range is the *binding* cause of imprecision. We test that implication directly with a renderer gate and a positive-control intervention, and **both refute it**.

Measurement: the FPV cannot encode metric range where the policy operates (but see Fig. 4 — this is a fixable artifact, and range is not what gates precision)

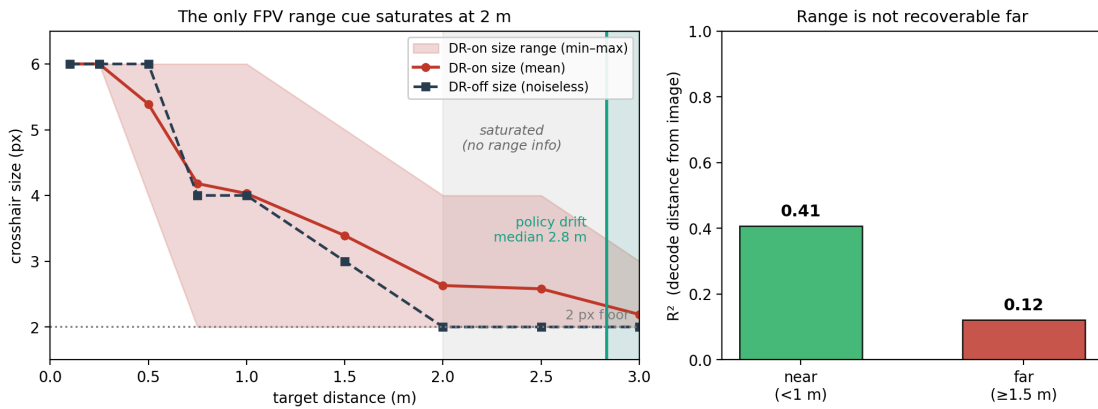


Figure 2. *Left*: the only FPV range cue (target crosshair size) saturates at the 2 px floor beyond ~ 2 m — under domain randomisation (red band = min–max) the size is uninformative even nearer, and with DR off (dashed) renders at 2.0/2.5/3.0 m are byte-identical. The policy's steady-state drift (green, median 2.83 m) sits inside the saturated region. *Right*: a linear decode of distance from the image succeeds near ($R^2=0.41$, < 1 m) but fails far ($R^2=0.12$, ≥ 1.5 m). This is a *measurement* of the observation, not the binding cause of imprecision (Figure 4).

The information loss is a renderer artifact, not the pixel count (gate). Before treating "richer sensing" as the fix, we ask whether a more capable monocular renderer would even *carry* the far-range information (`measure_higher_res_gate.py`; **Figure 4, left**). Decoding distance from the image across $\{64, 128, 256\}$ px $\times$ {production saturating crosshair, perspective non-saturating target}, the saturating crosshair's far-range R^2 is ≈ 0 at *every* resolution ($-0.01 / 0.11 / 0.04$) — raising resolution alone does nothing — whereas a perspective (optical-expansion) target lifts far R^2 to **0.42 at the same 64 px** ($0.50 / 0.45$ at 128 / 256 px). So the \$6.3 measurement is partly an artifact of this synthetic renderer's quantised crosshair, fixable without any resolution change; even so, the perspective target's far R^2 (0.45) $\ll$ its near R^2 (0.88), so a richer sensor would *improve*, not solve, far-range precision.

Even the oracle range cue does not move precision (intervention). The decisive test supplies the missing channel directly: we fold the metric body-frame position error the FPV cannot encode (`states[:, :3]`, computed from the existing data — no re-collection) into the policy's conditioning, retrain the DoE1 frontier recipe unchanged, and reveal under the frozen protocol (3 seeds; control = the no-cue DoE1; **Figure 4, right**). Handed the *oracle* metric range (scalar $\|\text{pos-err}\|$, $\sigma=0$), cond-IAE moves only $2.91 \rightarrow$ **2.43 m** — still $\sim 36\times$ the 0.068 m state oracle, and bought with -6.7 pp survival / -13 pp Tier-1. A realisable sensor ($\sigma=0.15$ m) erases even that (2.81 m $\approx$ control). The *richer* full 3-D position cue is actively harmful: it **collapses survival across all three seeds** (Tier-1 $92 \rightarrow 7\%$, surviving to the 250-step threshold in $\approx 2/30$ episodes), so its conditional precision is an artifact, not a win. Supplying range tells the drone *how far off* it is, but not *what to do* about it — and no demonstration in the 1–3 m band teaches that.

Verdict: precision is not sensing-gated. Range information — even oracle, even full position — does not restore precision, and a richer cue harms survival. The binding constraint is the absence of a learned far-range *recovery behaviour* in the 1–3 m band: the BC data has no labels there (coverage gap, above) and the teacher cannot generate them (the expert itself crashes from > 2 m, check 2). Moving precision would require a *competent far-range teacher* to

generate 1–3 m coverage — not a better sensor, not a better policy over the existing data, and not representation regularisation. A wider-init BC retrain is therefore still predicted to leave cond-IAE at ~ 2.8 m and is not pursued.

Precision is coverage/teacher-competence-gated, not sensing-gated

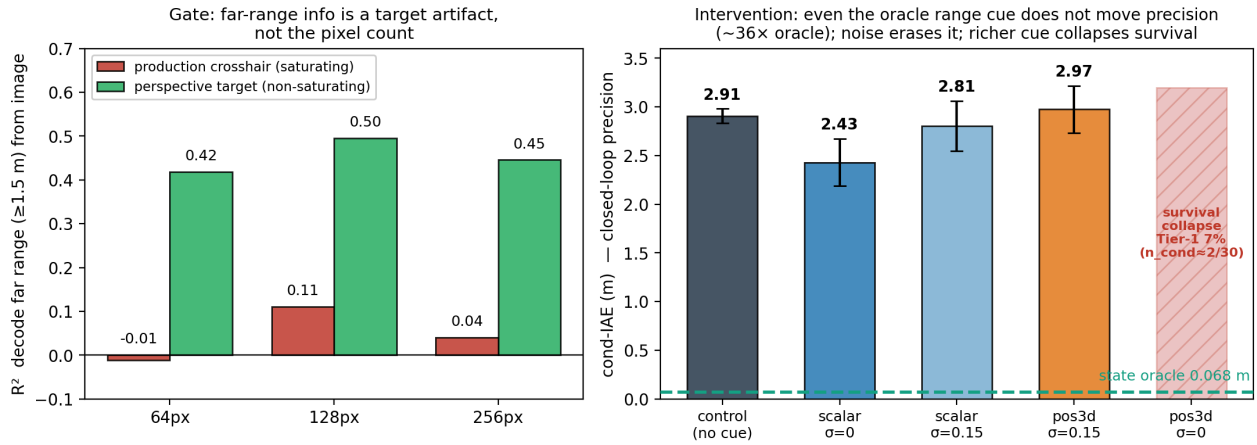


Figure 4. *Left (gate):* far-range (≥ 1.5 m) image→distance R^2 for three resolutions $\times$ two target renderers. The saturating production crosshair is uninformative at every resolution (raising pixels does nothing); a non-saturating perspective target restores $\sim$ half the far-range information already at 64 px — the loss is a target artifact, not the pixel count. *Right (intervention):* closed-loop precision (cond-IAE, 3 seeds) when the policy is handed the metric position error the FPV lacks. Even the oracle scalar range ($\sigma=0$) leaves precision $\sim 36\times$ the state oracle (green dashed, 0.068 m); sensor noise erases the gain; the richer 3-D cue ($\sigma=0$) collapses survival (hatched; conditional IAE unreliable). Precision is coverage/teacher-competence-gated, not sensing-gated.

7. Discussion

What generalises. (a) The frozen protocol and the short-survival artifact it fixes are reusable for any survival-vs-precision visual control benchmark. (b) The "objective-gaming" pathology of a *scale-sensitive* dispersion regulariser on an *off-path* feature is a general caution: a distance-maximising contrastive term can be trivially satisfied by norm inflation, *worsening* intrinsic rank; one should regularise the feature the policy actually conditions on, and use a scale-invariant criterion. (c) The rank-vs-survival decoupling argues that "prevent representation collapse" is not automatically the right lever for closed-loop control — the binding constraint must be located empirically.

What is specific. The precision diagnosis (§6.3) is specific to a setting whose privileged teacher saturates inside the operating regime: our state-based PPO oracle itself crashes from > 2 m offset, so it cannot label the 1–3 m band where the policy drifts. A task with a teacher competent over the full error range — or a renderer whose distance cue is not the quantised crosshair we found to be artifactually saturating — could place the bottleneck elsewhere.

Constructive implication. Improving metric precision here is neither a representation nor a sensing problem: our intervention shows that even the *oracle* range, handed directly to the policy, barely moves precision and a richer cue collapses survival (§6.3). The lever is the *teacher and its coverage* — a controller that recovers from multi-metre offsets, generating 1–3 m demonstrations to imitate. A better sensor would help only once such recovery behaviour exists to be conditioned on; absent it, higher resolution, stereo/depth, representation regularisation, and wider naive BC coverage are all predicted not to move the ~ 2.8 m floor.

8. Limitations

- **Simulation only;** no real-robot validation. The renderer is synthetic and its distance encoding is a design choice (a deliberately information-poor monocular cue) — though the §6.3 intervention controls for this by handing the policy the oracle range directly, so the precision conclusion does not hinge on the renderer's information content.

- The precision verdict is **conditional on the teacher**: our state-based PPO oracle itself crashes from > 2 m offset, so "coverage/teacher-competence-gated" is established for *this* teacher; a controller competent over the full error range is the untested lever and the recommended next step.
 - **One task** (hover) and one policy family (flow matching with IMU-vision cross- attention). The Dispersive null result is established for `vis_pooled` ; we did not exhaustively sweep λ , the feature it is applied to, or alternative dispersion criteria — though §6.1 explains mechanistically why those are unlikely to rescue it here.
 - $T_{\text{obs}} = 2$ frames could in principle yield range via motion parallax; we argue (and the flat empirical precision across all configs supports) that the 28 ms baseline and slow drift make this far weaker than the static size channel.
 - Three seeds per cell; effects are reported relative to the pooled across-seed std.
-

9. Conclusion

On a vision-based quadrotor hover task, under a frozen, artifact-robust protocol with a measured oracle, **Dispersive Loss does not improve survival or precision** above seed noise, and is a byte-identical no-op without an end-to-end encoder. The mechanism does not even cure the collapse it targets — it games its scale-sensitive objective and worsens intrinsic rank — and closed-loop survival is decoupled from that representation geometry entirely. The real levers are elsewhere: a transfer/conditioning/recovery recipe accounts for the survival gains, and **task precision is capped by the teacher's incompetence in the operating regime**, not by sensing: a positive-control intervention that hands the policy the oracle metric position error barely moves precision ($\sim 36\times$ the state oracle) and a richer cue collapses survival, and the FPV's apparent range-blindness is itself a fixable renderer artifact. Neither representation collapse nor the sensing channel is the binding constraint for this task; the missing far-range recovery behaviour is. We release the protocol, the ablation, the intervention, and all diagnostics so the result and its diagnosis are reproducible.

Reproducibility / Artifacts

Component	Script	Artifact
Frozen protocol + measured oracle	<code>scripts/evaluate_frozen_p0.py</code>	<code>evaluation_results/frozen_p0_leaderboard.json</code>
§4 P1 baselines (BC-vision-only, PPO-from-pixels), 3 seeds	<code>scripts/train_bc_vision_only.py</code> , <code>scripts/train_ppo_from_pixels.py</code> (+ <code>models/ppo_pixel.py</code> , <code>configs/ppo_from_pixels.yaml</code>), <code>scripts/evaluate_baselines_frozen.py</code> ; seed driver <code>scripts/run_baseline_seeds_1_2.sh</code>	seed 0: <code>evaluation_results/baselines_frozen_leaderboard.json</code> , ... <code>_ppopx.json</code> ; seeds 1–2: <code>..._{bc,ppopx}_s{1,2}.json</code> ; aggregate (mean±std): <code>..._baselines_frozen_seeds_aggregate.json</code>
2×2 ablation sweep / aggregate	<code>scripts/run_p2_ablation.py</code> , <code>scripts/evaluate_p2_ablation.py</code>	<code>evaluation_results/p2_ablation_{manifest,leaderboard}.json</code>
§6.1 feature geometry	<code>scripts/measure_feature_collapse.py</code>	<code>evaluation_results/p2_feature_collapse.json</code> — <code>docs/experiment_report_feature_collapse.md</code>
§6.2 survival movers	—	<code>docs/experiment_report_survival_movers.md</code>
§6.3 coverage probe	<code>scripts/measure_ood_coverage.py</code>	<code>evaluation_results/p3b_ood_coverage.json</code> — <code>docs/experiment_report_ood_coverage.md</code>
§6.3 image-distance info	<code>scripts/measure_image_distance_info.py</code>	<code>evaluation_results/p3b_image_distance_info{,_nodr}.json</code> — <code>docs/experiment_report_image_distance_info.md</code>
§6.3 higher-res gate (Fig 4 left)	<code>scripts/measure_higher_res_gate.py</code>	<code>evaluation_results/p3b_higher_res_gate.json</code> — <code>docs/experiment_report_sensing_ablation.md</code>
§6.3 range-cue intervention, 3 seeds (Fig 4 right)	<code>scripts/run_p3b_rangecue.py</code> (→ <code>train_flow_v5.py --range-cue</code> , <code>evaluate_frozen_p0.py --cue-noise</code>)	<code>evaluation_results/p3b_rc_{clean,noised}{,_s12}_frozen.json</code> — <code>docs/experiment_report_sensing_ablation.md</code>

Frozen protocol: 30 episodes, base seed 12345, $\sigma = 2.0$ exp-decay composite, paired init, conditional-IAE over episodes surviving $\geq 250/500$ steps, bootstrap 95% CI, measured PPO oracle 0.9668.

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